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A Implementation Details

A.1 Infrastructure

The experiments involved the 8 billion parameter model Llama-3.1-8B-Instruct and the 7 billion parameter model Qwen-2.5-7B-instruct. They were run on a single server with 8 NVIDIA RTX A6000 48 GB GPUs with CUDA Version 12.4 and an AMD EPYC 7413 24-Core Processor. The total runtime for training the probes and performing the grid search for steering was less than two weeks.

A.2 Libraries

See Table 6.

B Related Work on Grammatical Competence of LMs

Behavioral Evaluations of Grammatical Knowledge. Various benchmarks assess the linguistic knowledge of LLMs. BLiMP (Warstadt et al., 2020) and MultiBLiMP (Jumelet et al., 2025) evaluate syntactic acceptability via paired sentence probabilities, while HOLMES (Waldis et al., 2024) consolidates linguistic probing datasets across a range of syntactic phenomena, including tense classification tasks from Conneau et al. (2018) and Klafka and Ettinger (2020). Additional studies use probing methods to analyze specific grammatical categories, such as structural syntax (Hewitt and Manning, 2019; Diego Simon et al., 2024) and aspect in morphologically rich languages (Katinskaia and Yangarber, 2024). However, these studies typically model tense as a binary past–present distinction and do not address the full range of categorical tense and aspect distinctions.

Causal and Representational Analyses of Syntax in LMs. A complementary line of work investigates how grammatical information is encoded within LMs and how it can be manipulated. Studies have examined internal representations of syntax through attention patterns (Vig and Belinkov, 2019) and circuit-level structures (Wang et al., 2023; Ferrando and Costa-jussà, 2024), revealing how syntactic features are distributed across components of the model. Building on this, causal intervention methods have been used to identify which internal features are functionally relevant to grammatical behavior. CausalGym (Arora et al., 2024) tests whether linear representations influence syntactic decisions, while other work targets tense specifically, manipulating feed-forward layers (Merullo et al., 2024), attention heads (Zhang et al., 2025), or sparse autoencoder features (Brinkmann et al., 2025) to steer generation. These studies, like the behavioral evaluations, focus primarily on binary tense distinctions and are typically limited to single-token evaluations—with the exception of Brinkmann et al. (2025), who consider open-ended generation. Prior work has not jointly analyzed tense and aspect within a unified framework. Our work advances this area by studying categorical features in multi-token generation, combining probing, representation space analysis, and causal steering to examine how these concepts are encoded and can be controlled.

C Tense and Aspect Overview

We provide an overview of all possible tense-aspect combinations in the English language in Table 7.

Usage	Library	Model	Reference
Training Linear Probes	`scikit-learn`		Pedregosa et al. (2011)
Linear Discriminant Analysis	`scikit-learn`		Pedregosa et al. (2011)
POS-Tagging	`stanza`		Qi et al. (2020)
Propbank Annotations	`nltk`		Elhadad (2010)
	`spacy`	*en_core_web_lg*	Honnibal et al. (2020)
BERTScore	`bert_score`	*microsoft/deberta-xlarge-mnli*	Zhang et al. (2020)

Table 6: Libraries used for experiments.

	present	past	future
simple	She *drives* her car.	She *drove* her car.	She *will drive* her car.
progressive	She *is driving* her car.	She *was driving* her car.	She *will be driving* her car.
perfect	She *has driven* her car.	She *had driven* her car.	She *will have driven* her car.
perfect progressive	She *has been driving* her car.	She *had been driving* her car.	She *will have been driving* her car.

Table 7: Example sentence conjugated across different tense-aspect combinations.

D Dataset Composition

To ensure high-quality annotations for both tense and aspect, we prioritized careful dataset selection and manual validation. While existing resources such as TimeML (Pustejovsky et al., 2006) and Universal Dependencies offer valuable linguistic annotations, they were not an ideal fit for our specific needs, either due to differences in annotation focus or incomplete coverage of tense-aspect information. We also explored the TMV annotation tool (Ramm et al., 2017), but it relies on legacy part-of-speech (POS) and dependency parsing tools, which rendered it incompatible with our pipeline. Given these considerations, we curated a dataset with 692 sentences from PropBank (Palmer et al., 2005), each containing exactly one verb. To enforce this constraint, we applied dependency parsing and filtered out any sentence containing more than one word tagged with the POS label 'VERB'. While the filtering is not perfectly precise, it provides a reasonably effective heuristic for isolating single-verb constructions. To improve the overall data coverage for underrepresented tense-aspect combinations, we generated additional synthetic examples using ChatGPT-4o (Hurst et al., 2024), prompted with templates shown in the text box below. After downsampling the train set to address any remaining class imbalance, we have 348 samples per class for tense and 261 samples per class for aspect.

The test set from BIG-bench (Srivastava et al., 2023; Logeswaran et al., 2018) is balanced with ~70 samples per unique aspect and ~90 samples per unique tense.

Prompts for Synthetic Data Generation

Generate 100 diverse sentences in the [PAST] tense. Each sentence should contain only one verb and should vary in structure, subject, and length.

Generate 100 diverse sentences in [PAST] tense. Each sentence should not contain more than one verb and should vary in structure, subject, and length.

Generate a list of 100 random sentences that are in active or passive voice, declarative or interrogative, singular or plural. Each sentence should contain only one single verb phrase of the tense "[PAST]".

E Additional Probe Results

Besides length-normalized sum pooling, we explore naive summation, mean pooling, and the final token representation as alternative methods for extracting representations:

$$h_{\text{sum pooling}} = \sum_{i=1}^{N} h_i, \tag{7}$$

$$h_{\text{mean pooling}} = \frac{1}{N} \sum_{i=1}^{N} h_i, \tag{8}$$

$$h_{\text{final token}} = h_{-1}. \tag{9}$$

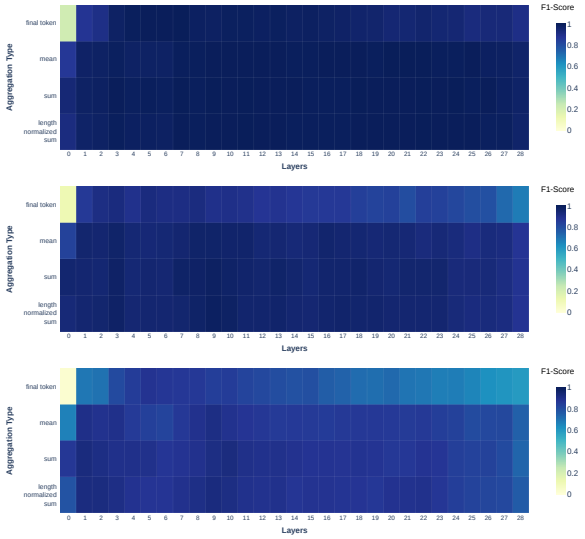

Figure 7: Qwen-7B probing f1-scores for **tense**, **aspect** and **tense-aspect** across layers.

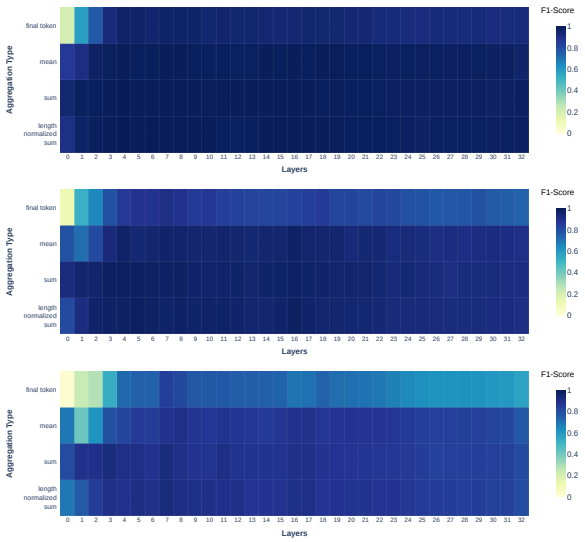

Figure 6: Llama-8B probing f1-scores for **tense**, **aspect** and **tense-aspect** across layers (L0: embedding layer).

We provide layer-wise probing results across different aggregation strategies in Figure 6 for Llama-8B and in Figure 7 for Qwen-7B. Across both, models and targets, sum pooling and its length normalized version yield the highest f1-scores. Using the final token representation only is slightly less informative and limited to the earlier middle layers.

F Cluster Quality of LDA Projections

To quantify the separation effectiveness of our LDA directions, we compute the following metrics for the L0 projected hidden states and report them in Table 8 (visualizations of Qwen-7B: Fig. 1, 2):

- **Explained Variance (↑):** ratio of between-class variance and total variance
- **Fisher Discriminant Ratio (↑):** between-class vs within-class variance
- **Silhouette score (↑):** similarity of a point to its own class vs others.

Further, to measure the orthogonality of the parent feature directions $\bar{\ell}_{\text{TENSE}}$ and $\bar{\ell}_{\text{ASPECT}}$, we compute the cosine similarity between all possible vector differences for tense and aspect respectively, considering the layers with the best steering efficacy across tasks and methods. This results in a mean similarity of 0.045 for Llama-8B and 0.118 for Qwen-7B. Cosine values this small imply that the two contrast directions are almost orthogonal, so the models largely encode tense and aspect in independent subspaces.

G Tasks for Multi-Token Steering

We consider three generative tasks to evaluate the steering procedure, their prompt formats are detailed below.

Random Sentence Task

Using the template "`<Imperative Verb> <Sentence Description>:`" with values from below, we form a test set of $N = 83$ distinct prompts for the random sentence task (e.g., "Formulate one grammatically correct sentence:").

Imperative Verbs
Generate, Create, Produce, Write, Output, Provide, Construct, Make up, Formulate, Come up, Print, Return, Craft

Sentence Descriptions
a single sentence, one sentence, a random sentence, a sentence using any verb tense, an arbitrary sentence, one grammatically correct sentence

For repetition and temporal translation, we only include samples where the unsteered output is a valid answer.